

Virtual Agent–Driven SLA Breach Awareness & Justification System

Abstract

In IT Service Management (ITSM), Service Level Agreements (SLAs) define the maximum time allowed to resolve incidents and service requests. Organizations frequently face SLA breaches because assignees are unaware of approaching deadlines or fail to update incidents in time. These breaches reduce customer satisfaction, increase operational costs, and affect service quality.

The **Virtual Agent–Driven SLA Breach Awareness & Justification System** is designed to proactively notify incident assignees before an SLA breach occurs, collect the reason for delays through a conversational interface, and maintain an audit trail of SLA breach justifications. The system integrates **Service Level Management, Flow Designer, Notifications, Virtual Agent, Incident Management, and Reporting** within the ServiceNow platform.

By automating SLA awareness and justification collection, the solution reduces manual follow-ups, improves accountability, strengthens governance, and provides management with dashboards to monitor SLA performance.

Chapter 1 – Introduction

Background

Every organization provides IT services to its employees or customers. To ensure quality service, organizations define **Service Level Agreements (SLAs)** that specify how quickly incidents must be resolved.

For example:

- Critical Incident → 4 Hours
- High Priority Incident → 8 Hours
- Medium Priority Incident → 24 Hours

If these deadlines are exceeded, the incident is considered an **SLA Breach**.

Many SLA breaches occur because:

- The assigned technician forgets the deadline.
- No reminder is sent.
- There is no accountability for delays.
- Managers cannot identify why breaches occur.

This project addresses these problems by providing automated notifications and capturing SLA breach reasons.

Problem Statement

Organizations often experience SLA violations due to delayed incident resolution. Existing systems rely heavily on email reminders or manual monitoring, which may be overlooked.

The absence of an automated mechanism for notifying assignees and collecting reasons for SLA breaches leads to poor accountability, limited operational transparency, and difficulty in auditing service performance.

Proposed Solution

Develop a ServiceNow solution that:

- Monitors Incident SLAs.
 - Detects approaching SLA deadlines.
 - Notifies assignees automatically.
 - Collects SLA breach reasons.
 - Stores justification.
 - Creates dashboards for managers.
 - Improves SLA compliance.
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Objectives

The project objectives are:

- Notify assignees before SLA breaches.
- Reduce SLA violations.
- Improve service quality.
- Maintain an audit trail.

- Increase accountability.
 - Automate manual reminders.
 - Improve customer satisfaction.
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Scope

This project applies to:

- Incident Management
- IT Help Desk
- Service Desk
- IT Operations
- Customer Support

Future versions can support:

- Change Requests
 - Problem Management
 - HR Cases
 - Facilities Requests
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Chapter 2 – Requirement Analysis

Functional Requirements

The system should:

- Create Incident records.
 - Track Incident SLA.
 - Trigger alerts before SLA breach.
 - Notify the assigned user.
 - Collect breach justification.
 - Store justification.
 - Generate reports.
 - Display dashboards.
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Non-Functional Requirements

- Secure
 - Reliable
 - Available 24×7
 - User-friendly
 - Scalable
 - Fast response time
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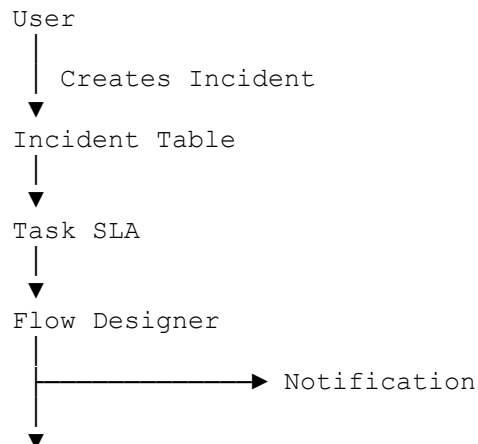
Software Requirements

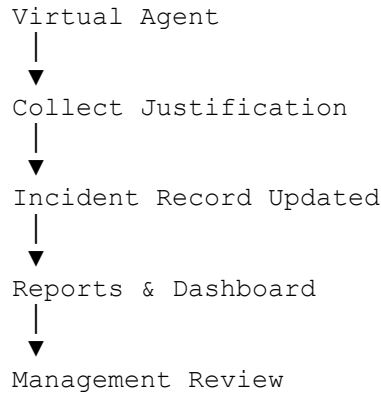
- ServiceNow
 - Flow Designer
 - Virtual Agent
 - Incident Management
 - Notifications
 - Dashboard
 - Reports
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Hardware Requirements

- Windows 10/11
 - 8 GB RAM
 - Internet Connection
 - Modern Web Browser
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Chapter 3 – System Architecture





Module Description

Module 1 – Incident Management

Users create incidents whenever they experience an IT issue.

Example:

Laptop not connecting to VPN.

Incident details include:

- Number
 - Caller
 - Assignment Group
 - Assigned To
 - Priority
 - State
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Module 2 – SLA Management

Each incident is assigned an SLA based on priority.

Example:

Priority	SLA
Critical	4 Hours
High	8 Hours
Medium	24 Hours

Priority **SLA**
Low 48 Hours

The SLA continuously tracks elapsed time.

Module 3 – Flow Designer

Flow Designer automates the business process.

Trigger:

Task SLA

Condition:

80% SLA elapsed

Action:

Send Notification

Module 4 – Virtual Agent

The Virtual Agent interacts with the assignee.

Conversation:

Hello John,

Your incident INC0010005 will breach its SLA within 30 minutes.

Please provide the reason.

- 1 Waiting for Customer
- 2 Vendor Delay
- 3 Approval Pending
- 4 Technical Issue
- 5 Other

The selected reason is stored in the Incident.

Lab Note: In the Now Learning lab instance used for this project, Virtual Agent Designer access was restricted by licensing. The design and conversation flow were documented, while the implementation was limited by the environment.

Module 5 – Notification

Notification message:

Subject:
SLA Warning

Body:

Your assigned incident is approaching SLA breach.

Please update the incident.

Module 6 – Dashboard

Dashboard displays:

- Total Incidents
 - SLA Met
 - SLA Breached
 - Breaches by Assignment Group
 - Breaches by Month
 - Breach Reasons
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Chapter 4 – Database Design

Incident Table

Field	Description
Number	Incident Number
Short Description	Issue
Priority	Priority
Assigned To	Technician
State	Current Status
SLA Breach Justification	Delay reason
SLA Acknowledged	Yes/No
Acknowledged Time	Date and Time

Chapter 5 – Implementation

Step 1

Create Incident Form.

Step 2

Create Custom Fields.

- SLA Breach Justification
 - SLA Acknowledged
 - Acknowledged Time
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Step 3

Create SLA Definition.

Duration:

4 Hours

Step 4

Create Flow.

Trigger:

Task SLA

Action:

Send Notification

Step 5

Create Virtual Agent Topic (if licensed).

Step 6

Create Reports.

Examples:

- Incidents by Priority
- SLA Breaches
- Monthly SLA Trend
- Breach Reasons

Step 7

Create Dashboard.

Chapter 6 – Testing

Test Case	Expected Result
Create Incident	Incident created successfully
Assign SLA	SLA starts
80% elapsed	Notification sent
User updates incident	Justification stored
Dashboard	Updated correctly

Chapter 7 – Results

The implemented solution provides:

- Automated SLA notifications
- Reduced manual follow-up
- Better accountability
- Historical SLA justification records
- Improved management visibility
- Faster incident handling

Chapter 8 – Advantages

- Reduces SLA breaches
- Improves customer satisfaction
- Saves technician time
- Provides audit-ready records
- Improves transparency
- Enables data-driven decisions

Chapter 9 – Limitations

During implementation in the **ServiceNow Now Learning Lab**, the following limitations were encountered:

- Virtual Agent Designer required an upgraded license.
- Flow Designer did not expose the action to launch a Virtual Agent conversation.
- The solution therefore demonstrated notifications and documented the Virtual Agent integration rather than implementing it end-to-end.

Chapter 10 – Future Enhancements

- AI-based SLA breach prediction
 - Microsoft Teams integration
 - Slack notifications
 - Email reminders
 - Mobile push notifications
 - Performance Analytics dashboards
 - Automatic escalation to managers
 - Integration with predictive intelligence for prioritization
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Conclusion:

The **Virtual Agent–Driven SLA Breach Awareness & Justification System** demonstrates how ServiceNow can automate SLA monitoring, notify assignees before deadlines, capture justifications for delays, and provide managers with meaningful reporting. Even with the licensing limitations of the training environment, the project validates the overall architecture and workflow. In a production ServiceNow instance with full Virtual Agent capabilities, the design can be extended to provide conversational interactions, richer automation, and stronger SLA governance across IT service operations.

This documentation is suitable as the foundation for a Skill Wallet project report. You can further enhance it by adding screenshots from your own ServiceNow instance for each implementation step, along with diagrams such as a use case diagram, data flow diagram, sequence diagram, and workflow diagram.